07 Provenance

Cryptographic Provenance Benchmark — ESPHOME

Company: Anticloud FZ LLE

Receipt Generation Performance

Operation Time
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Single receipt creation 0.8ms
Chain verify (100 entries) 2.1ms
Chain verify (10K entries) 198ms
File hash (1MB) 4.2ms
File hash (1GB) 4,200ms

Chain Integrity Tests

Test Result
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Sequential append PASS
Tamper detection (modify entry 50 of 100) PASS — detected at entry 50
Reload from disk PASS — head matches
Cross-node receipt verification PASS — node fingerprint stable

Hash Algorithm

SHA3-256 (NIST FIPS 202). Collision resistance: 2^128.
No MD5, no SHA1, no SHA2-256 (SHA3 only).